

SIMATS ENGINEERING

COURSE CODE: CSA1412

ASSESSMENT TOOL 2

COURSE NAME: COMPILER DESIGN

COURSE OUTCOME COVERED

CO1: *Analyze* syntax-directed translation method *using* synthesized *and* inherited attributes *to* generate a Parser Tree. (BL4)

Assessment Tool Weightage: *20%*

QUESTIONS: 5

Annexure A

CASE STUDY

Duration :60 Mins On Class

Total Marks:50

Code of Conduct:

I Mathusri P-192311363 (Name/Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules

will result in disciplinary action.

Signature of the student: Mathusri P

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CSA1412-Compiler *Design*

CO3. AT2. Case study
assignment

• Syntax Tree and SDD for

$a + b * c - d.$

Syntax *Tree* Given: ~-~

m

$a + b * c - d$

procedence:

m

Mathushi P

192311363

- < + < *

*

Associativity.

$+$ and $-$ *are* left
associative

Therefore, the
expression is
interpreted

$(a + cb * c) - d$.

Syntax Free:

a * **^**

b c

+ **d**'

as :-

Thus multiplication
is performed first:
followed by addin

and finally
subtraction.

(i) (Syntax –
Directed Definition
(SDD).

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$E \Rightarrow E + T \mid E - E - T \mid E - T$

$T \rightarrow T * F$

$T \rightarrow F$

$F \rightarrow id$

semantic rule:

$E \rightarrow E1 + T$

$E \cdot \text{val} \Rightarrow E_{\text{val}} + T_{\text{val}}$

$E - E_i - T$

$E.\text{val} \Rightarrow E_i.\text{val} + T_{\text{val}}$

$E \rightarrow T$

$E_{\text{val}} = T_{\text{val}}$

$1 * 114 - 1$

$T_{\text{val}} = T_i.\text{val} F_{\text{val}}$

$T - F$

$T.\text{val} \Rightarrow F.\text{val}$

$F - id$

F.valid.lexval.

First:

$b * c \Rightarrow 3 \times 4 \Rightarrow 12 //$

Then,

$a + 12 \Rightarrow 72 + 12 = 14 / 1$

Finally,

$14 - d \Rightarrow 714 - 5 \Rightarrow 79 //$

Attribute propagation:

..all *val* attributes use
synthesized

* Attribute Evaluation:
attributes.

Liii)

for,

$a + b * c - d$

$b.val = 3$ $C.val = 4$

↓

$Toval = boval * C.val$

$= 12$

$9.val = 2$

↓

$E.val = 9.val + T.val$

1

$= 14$

`d.val = 5`

↓

`Eval = 14-5`

(iv) Order of Operations :)

The syntax tree itself
determines the evaluation

Loader

a

+ d

*

assume;

$A = 2$

$b = 3$

$C = 4$

$= 5$

b

c

The deepest operator u,

10. it is evaluated first

$b * c$

Then: $a + Cb * c$)

Finally: $C(a + Cb * c)) - d$

Result:

Expression : $a + b + c - d$

* structure: $ca + cb^*c) - d$

* precedence: $* > + > -$

→ associativity: left

The

lyntax

ryntore tree correctly

represents the precedence

& left associativity of the
expression.

S-Attributed SDD and Bottom-up
Translation.

Given Grammar.)

Grammer, HA 2

S-En

EE+T

IE-T

IT

TT*F

1T/F

IF

F-(E)

ldigit

i/p: $8+2*4n...$

S-attributed SDD

use synthesized attributerval.

S-En

S.val = Eval.

$E \rightarrow E_1 + T$

E.val = E₁.val + T.val.

$E \rightarrow E_1 \cdot T$

E.val = E₁.val - T.val

$E \rightarrow T$

E.val = T.val

$T \rightarrow T_1 F$

T₁.val = T₁.val * F.val

$T \rightarrow T_1 / F$

T₁.val = T₁.val / F.val

T-F

T.val = F.val

F → CE)

4

F.val = Eval

F → digit

F.val = digit · lexical.

Translator/code

Fragment

a simple bottom up translator is:

E-EIT

emit ("+", E1.val, T-val)

$T \rightarrow TI * F$

`emit ("*", Ti.val, F.val)`

$F \rightarrow \text{digit}$

`Eval = digit · lexval.`

2

2

(ii) Parser stack Evaluation

i/p : 8+ 3* un

Stack

Action.

8

Shift 8

(4) [16]

8[8]

(*) [8]

2[2]

4 [4]

LL

Reduce $F \rightarrow \text{digit}$

T

Reduce $T \rightarrow F$, value $\rightarrow 8$

E+

shift +

E+2

Shift 2

E+T

Reduce 2 to T

E+T*

shift*

shift 4

$E+T*4$

$E+T*F$

Reduce F-sdigit,
(V) Precedence and
Associativity

The grammar separates:-

$E \rightarrow E+T$

$T \rightarrow T*F$

... T is processed before E

hence

$8+2*4$

↓

$$8+8$$

value = 4.

E+T

Reduce $T \rightarrow T * F$

E

Reduce $E \rightarrow ETT$

co

Reduce $S \rightarrow E_n$

intermediate calculations:

$$2 \times 4 = 8$$

$$8 + 8 = 16$$

(1) Annotated Syntar Tree.

↓

16.

S- Attributed L-Attributed and
circular Dependencies

Production

$A \rightarrow BCD$

Each non-terminal has:

S=

S- synthesized

i = inherited.

H

As B.itc-i

-

S-attributed? No

Reason: B.i and C.i are
inherited attributes

B.i and C.i are inherited
attribute. Input: real p, q, r

L-attributed?

The rule does not show how B.i and
C.i are defined. Therefore
it is not a complete evaluable
L-attributed SDD.

Conclusion:.

S-attributed: No

L-attributed : Not completed/
can't be fully evaluated.

(H) Evaluation:

$$2*4=8$$

$$8+8=16$$

(i) Set S-attributed attributes
Circular

$$\begin{matrix} 01 \\ 1:= \\ = \end{matrix}$$

No

(4

SDD

$D \rightarrow TL$

$L.in = T.type$

$T \rightarrow real$

$T.Type = real$

$L \rightarrow L, id$

$L.in = L.in$

$id.type = L.in$

$L.id$

$id.type = L.in$ (ii) Attributes:

$T.type = Synthesized$

$L.in = Inherited$

id-type = stores the
identifier's type.

(ID) (Evaluation)

real

$A.S = B.i + C.S$

$Doi = A.i + B.S$

for A-BOD

L-

No

No

To type = real

Yes

No

$L_{\text{in}} = \text{real}$

Yes

No

$9.\text{type} = \text{real}$

Yes.

$r.\text{type} = \text{real}.$

iv

No

Yes

No

مله

(iv) $A.s \rightarrow B.i$ Di-As

This Creates a circulous
Dependary

There fore. is No valid evaluation
urde,

↓

P.type = real

(iv) Annotated Tree.

real

P

D

T (real) i

Lin = real

2
.Y

real real.

(V) Result.

D- real

9- real

r-real

Thus, The type **real** is correctly
Propagated to all identifier.

C (input: $3*4 + 61$)

(1) L- Attributed SDD

ETE'

$$E'.inh = T.val$$

$$E.val = E'.Syn$$

$$E' \rightarrow +TE'I$$

$$E'.inh = E'.inh + T.val$$

$$E' - \varepsilon$$

$$E' - Syn - E', inh$$

$$'T-FT'$$

$$Tinh - Foval$$

$$Tival T'syn$$

$$T-7*FT'I$$

$$T'1-inh = T' inb * Foval.$$

$$T'-E$$

$T'.syn = T.inh$

F-digit

$F.val = digit.lexual.$

(1) (Attribute Evaluation

$3 \rightarrow F.val = 3$

$4 \rightarrow Foval = 4$

$3 * 4 = 12 \quad 12 + 6 = 18$

(ii) Annotated Tree.

$+ [18]$

$* [12] \quad 6$

$3 \quad 4$

(instinal Anuver:

Exprenton value =
18

=18

multiplication is performed
before addition so, the required
operator precedence is
maintained